in the future to support the open source NVIDIA kernel driver as an alternative to the Nouveau DRM driver. The original NVK open source Vulkan driver code is available on Nouveau's GitLab repository.

Spotify launches 'Basic Pitch', an audio-to-MIDI file converter

The R&D team at Spotify has released Basic Pitch, a free, open source tool for converting audio files to MIDI format. Basic Pitch, which uses machine learning to transcribe musical notes, can convert a recording of almost any instrument — including voice — to a MIDI version. Over the last 40 years, musicians who use computers to compose, produce, and perform their music have primarily used MIDI, a digital standard that acts as sheet music for computers, describing which notes to play when in an easy-to-edit format. Despite its widespread use, creating a composition from scratch with MIDI can be difficult. MIDI notes are typically generated by musicians using an interface, such as a keyboard designed for that purpose, or by manually typing the notes into their software.

Live performances with real instruments are frequently difficult for a computer to interpret, which is a problem for singers who are unfamiliar with piano keyboards or complex software. Even for musicians who are familiar with MIDI composition, it can be a time-consuming process.

"To solve this problem, researchers at Spotify's Audio Intelligence Lab teamed up with our friends at Soundtrap to build Basic Pitch — a machine learning model that turns a variety of instrumental performances into MIDI," said a company release.

Basic Pitch has advantages over other note-detection systems, such as pitch bend detection and tracking multiple notes across multiple instruments at the same time. It is also low on resources, allowing it to run faster than other systems of its type on most modern computers. Basic Pitch aims to provide musicians and producers with the "power and flexibility" of MIDI without the need for specialised equipment, allowing them to record whenever inspiration strikes and edit their compositions later.

Keyfactor unveils its community

Keyfactor, the machine and IoT identity platform for modern enterprises, has launched the Keyfactor Community. The community will provide knowledge and open source tools to developers, and operations and engineering teams to help them implement the best security solutions for their use case or product.

To build, deliver, and run applications securely, engineering and operations teams are increasingly relying on public key infrastructure (PKI) and machine identities. PKI provides developers

JFrog announces Pyrsia to secure software

JFrog Ltd, the liquid software company that created the JFrog DevOps platform, has announced Project Pyrsia, an open source software community initiative that uses blockchain technology to secure software packages from vulnerabilities and malicious code. Project Pyrsia, which is now accepting sign-ups, is an open source based, decentralised, secure, build network and software package repository aimed at assisting developers in establishing a chain of provenance for their software components, thereby increasing confidence and trust.

Open source software is an essential component of nearly every technology we use today. Nonetheless, there is no doubt that the volume, sophistication, and severity of software supply chain attacks have increased in the last year. The JFrog Security Research team has tracked over 20 different open source software supply chain attacks in recent months, two of which were zero-day threats. While open source components are intended to improve development efficiency, not knowing where your software comes from makes it difficult to identify risks, sowing doubt and uncertainty about its safety.

As a result, JFrog and other open source technology leaders such as Docker, DeployHub, Futureway, and Oracle have collaborated to create the Project Pyrsia network for validating the source and security of open source software packages. Pyrsia enables developers to use open source software with confidence, knowing that their components have not been compromised, without the need to build, maintain, or operate complex processes for securely managing dependencies.